GROUP : test

06026225 DEEP LEARNING IN MEDICAL IMAGE AND VIDEO ANALYSIS

THAI CHARACTER & NUMBER RECOGNITION

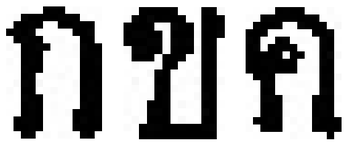


66070203 | 67070112 | 67070147 | 67070158 | 67070201 | 67070291

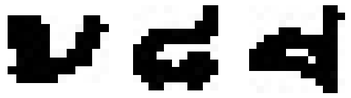
RECAP



72 CLASSES



(CONSONANTS)

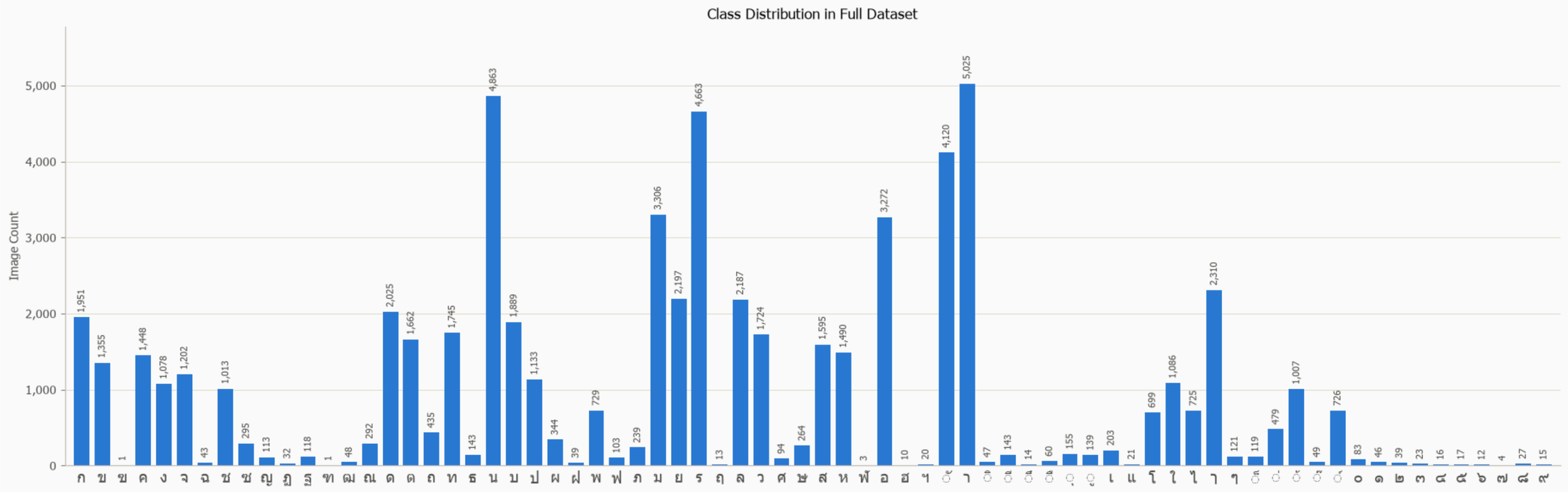


(VOWELS & TONES)



(NUMBERS)

PROBLEMS



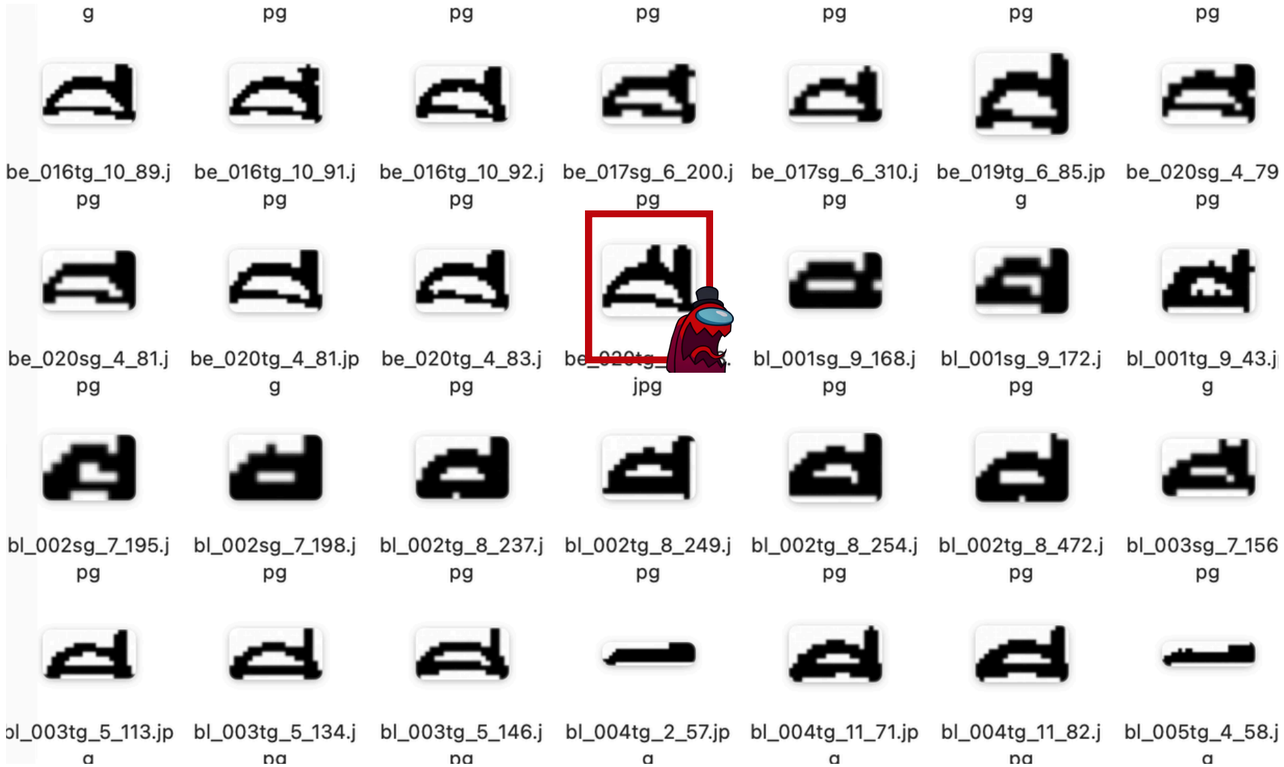
CLASS IMBALANCE

PROBLEMS



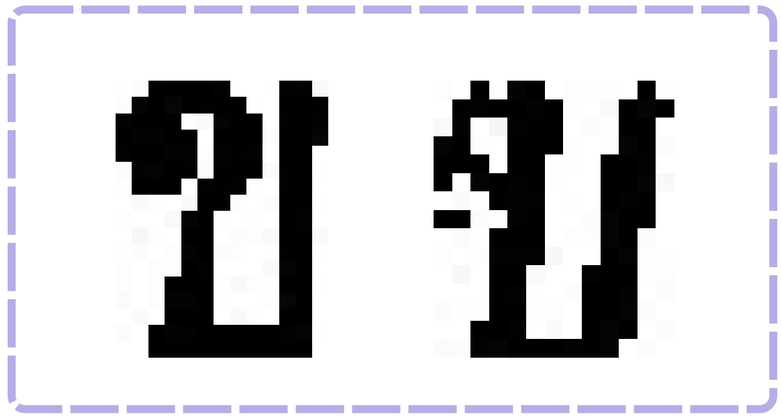
RARE CLASS

PROBLEMS



CONTAMINATED DATA

PROBLEMS

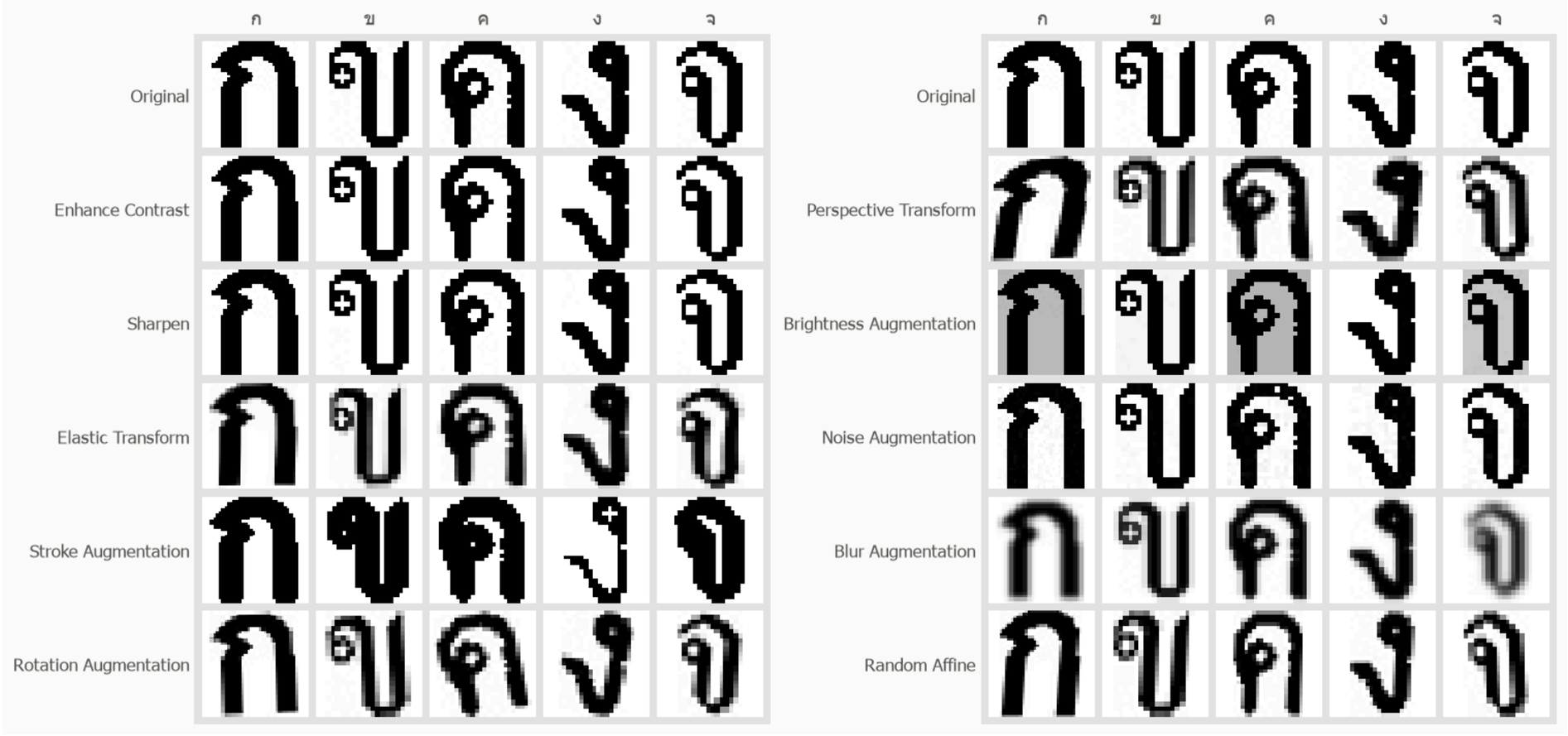




VISUAL SIMILARITY

METHODOLOGY

- DATA AUGMENTATION



METHODOLOGY

- DATA AUGMENTATION

NEW CLASS + 8



METHODOLOGY

- TRAIN TEST SPLIT

TRAIN 80%

TEST 20%

80-20

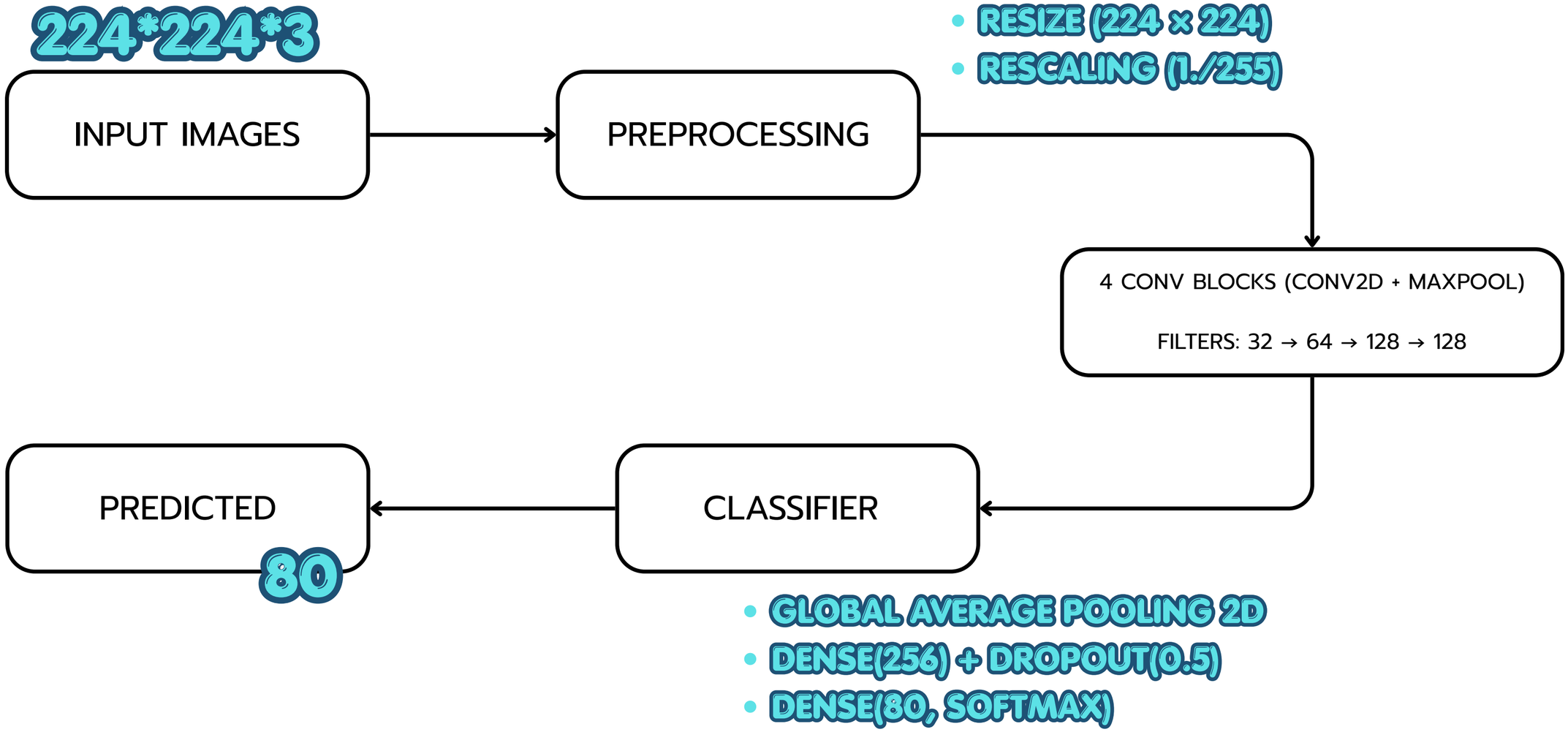
TRAIN 70%

VALIDATION 15%

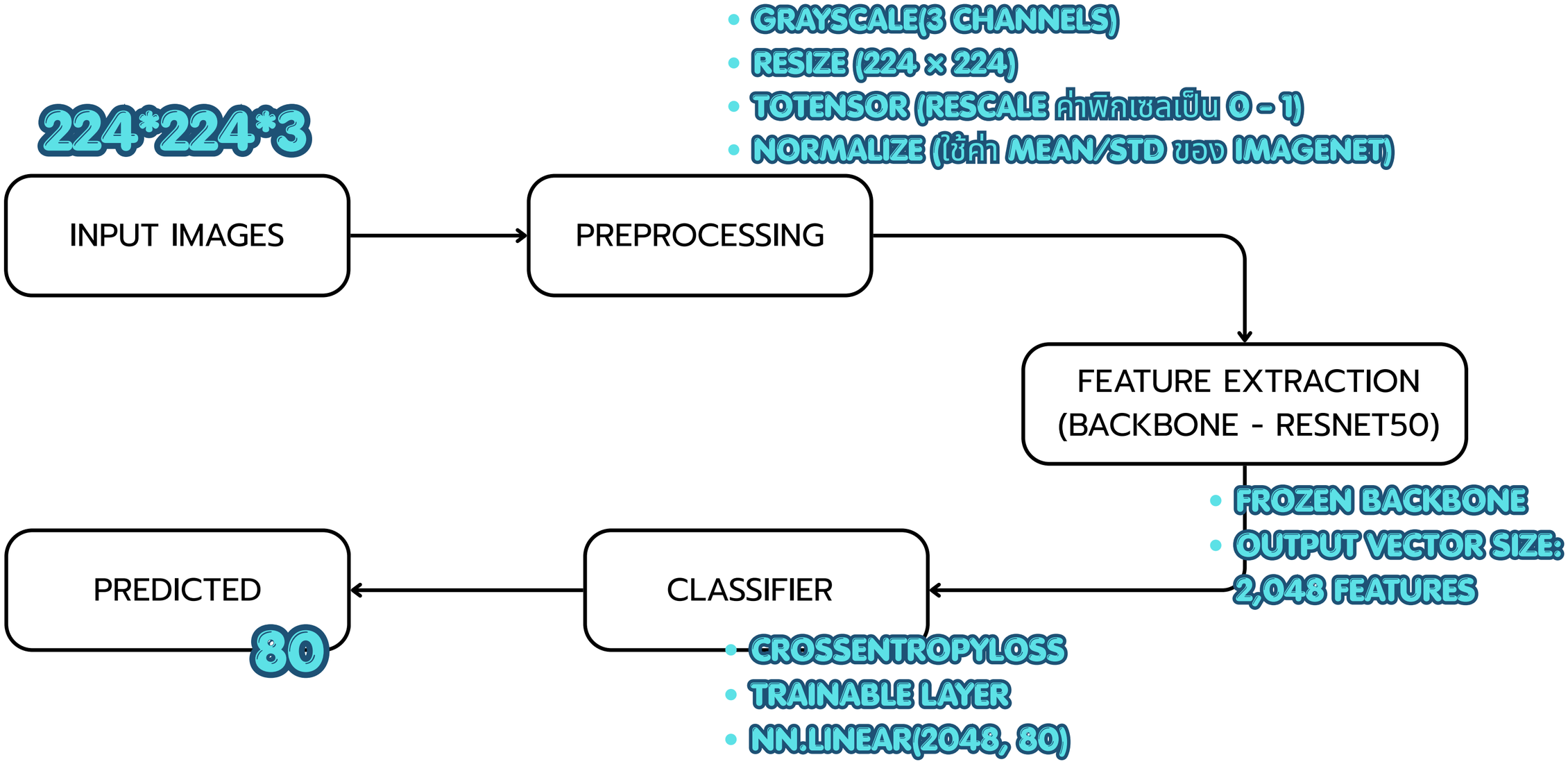
TEST 15%

70-15-15

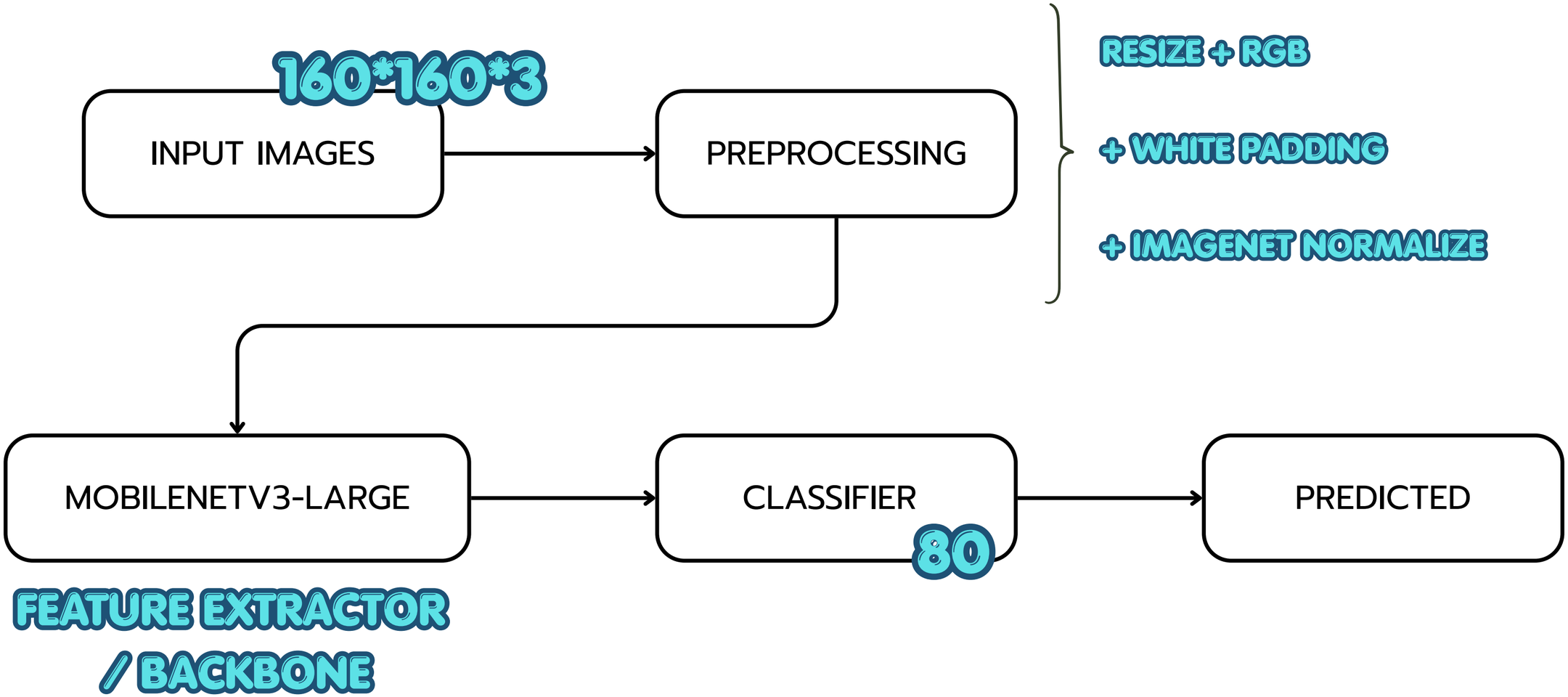
ARCHITECTURE — CNN



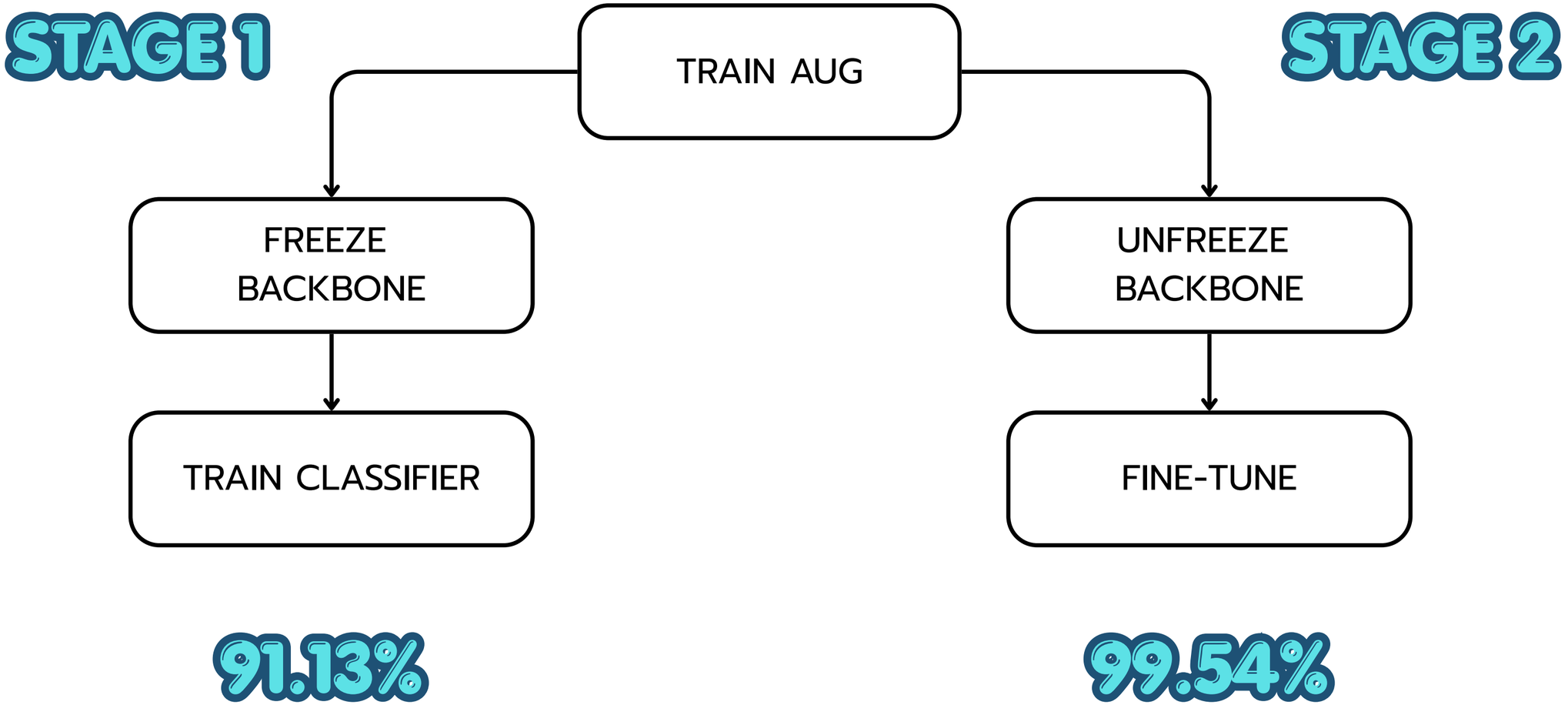
ACHITECTURE — RESNET50



ACHITECTURE — MOBILENETV3-LARGE



TRANSFER LEARNING STRATEGY — MOBILENETV3-LARGE



TRAINING TECHNIQUES — MOBILENETV3-LARGE

CLASS-WEIGHTED CROSS ENTROPY

รับมือ CLASS IMBALANCE

LABEL SMOOTHING

ลด OVERCONFIDENCE/ OVERFITTING

MIXED PRECISION

ลด VRAM และเพิ่มความเร็ว

GRADIENT CLIPPING

ทำให้ GRADIENT ไม่รุนแรงเกินไป

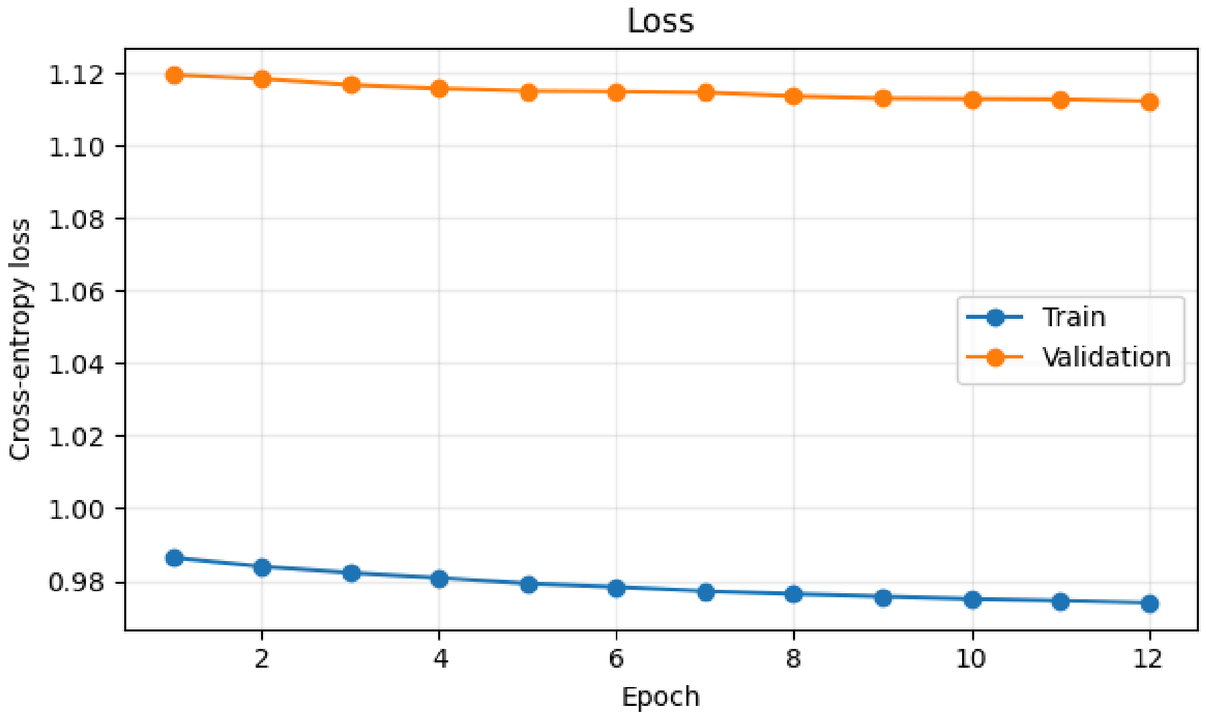
REDUCELRONPLATEAU

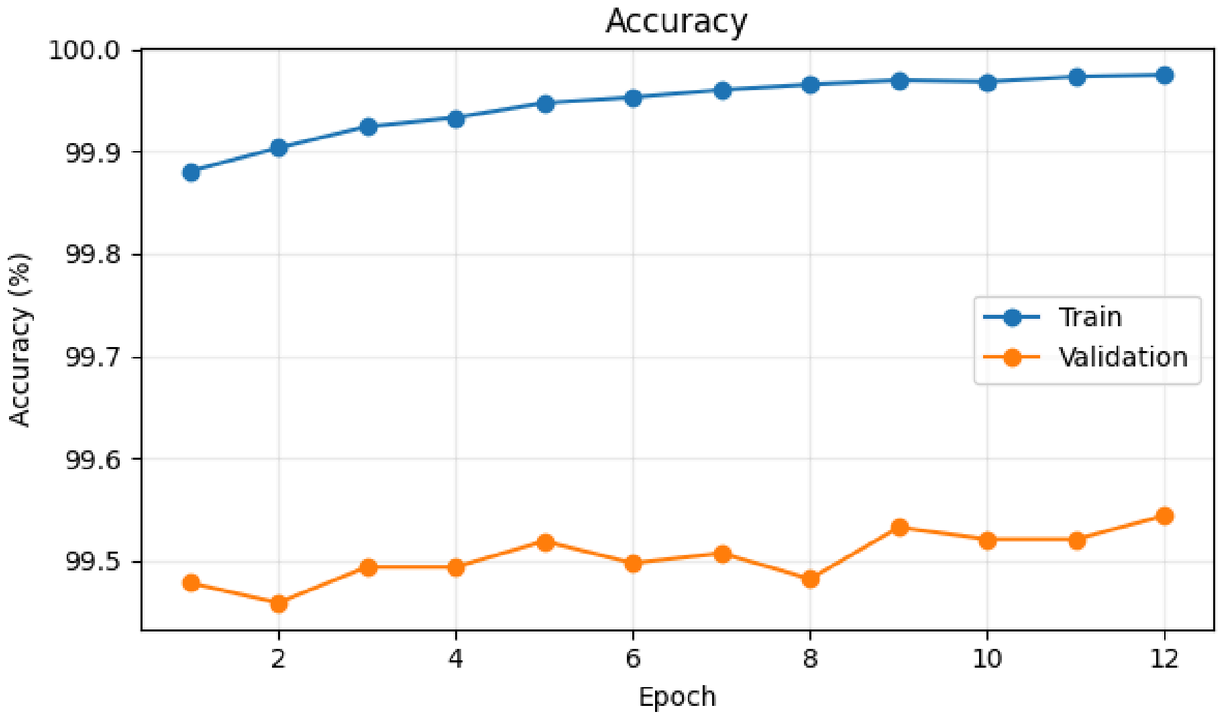
ปรับ LEARNING RATE เมื่อ VALIDATION ไม่ดีขึ้น

EARLY STOPPING

หยุด TRAINING เมื่อโมเดลไม่ดีขึ้น

LEARNING CURVES — MOBILENETV3-LARGE

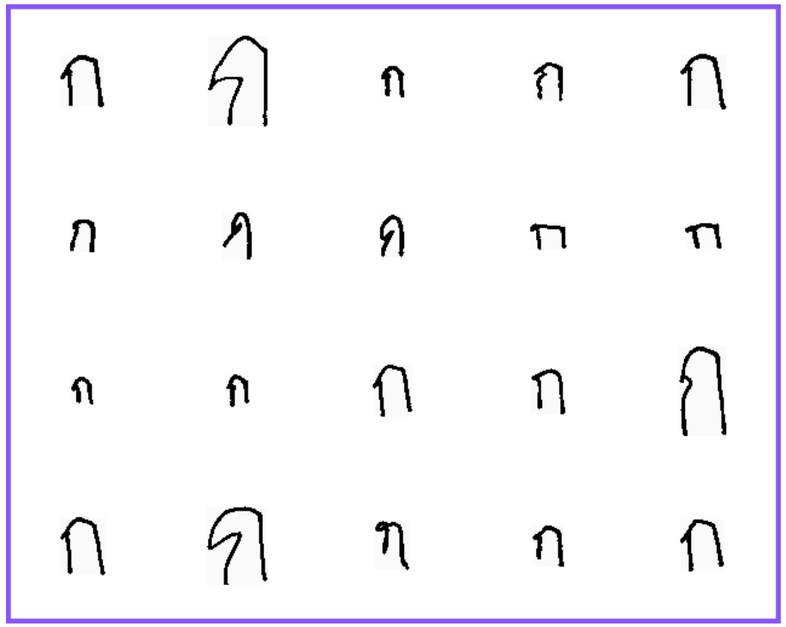




EVALUATE

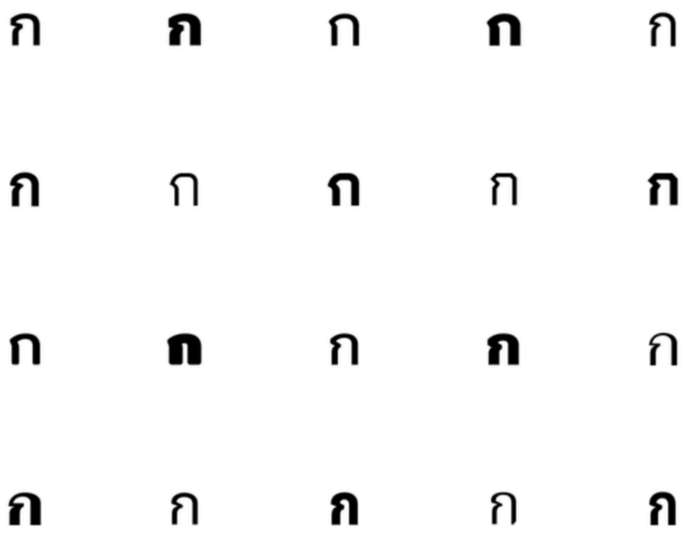
ALICE-THI

7,700 PIC
77 CLASS



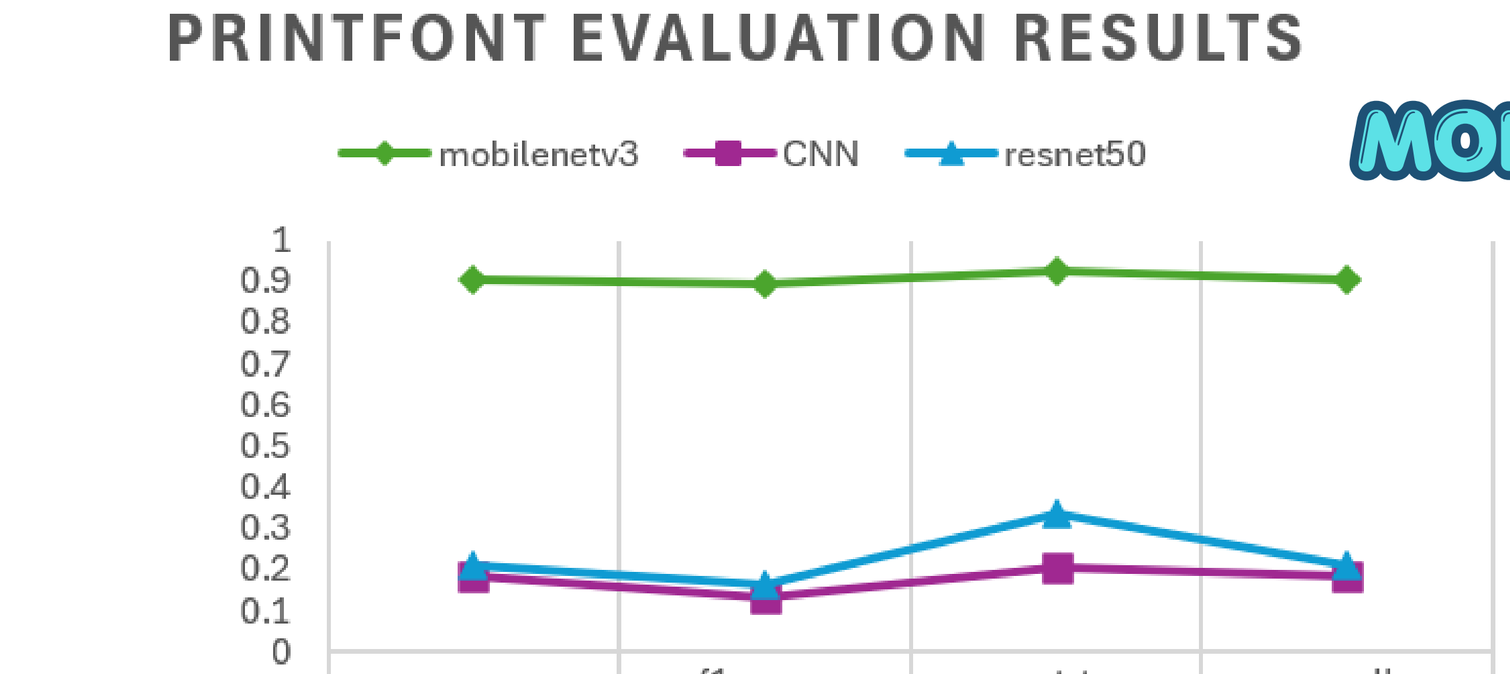
PRINTFONT

32,000 PIC
80 CLASS



https://www.ai.rug.nl/~mrolarik/ALICE-THI/

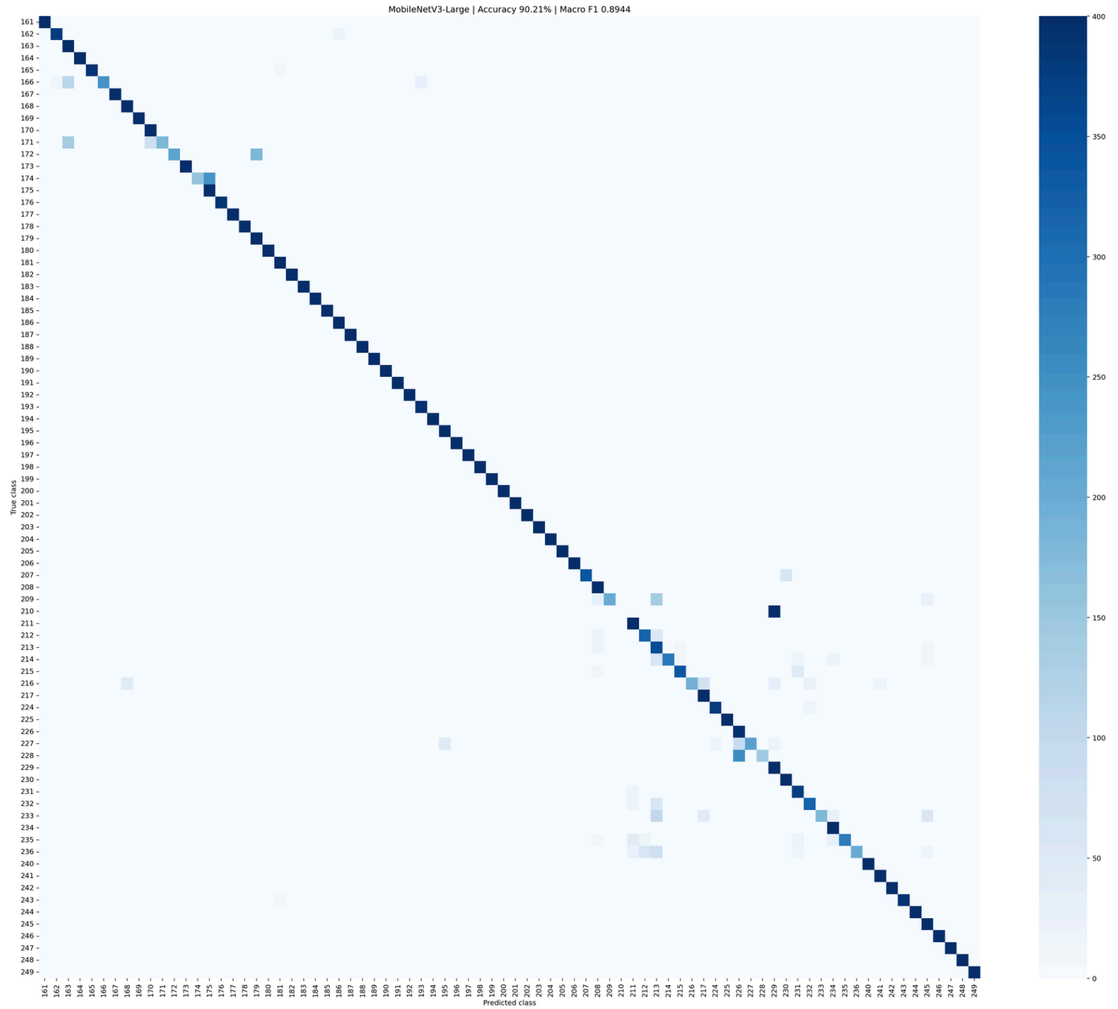
EVALUATE



	accuracy	f1-score	precision	recall
mobilenetv3	0.9021	0.8944	0.923	0.9021
CNN	0.1846	0.1302	0.2036	0.1846
resnet50	0.2094	0.1635	0.3351	0.2094

MOBILENETV3
ดีที่สุด

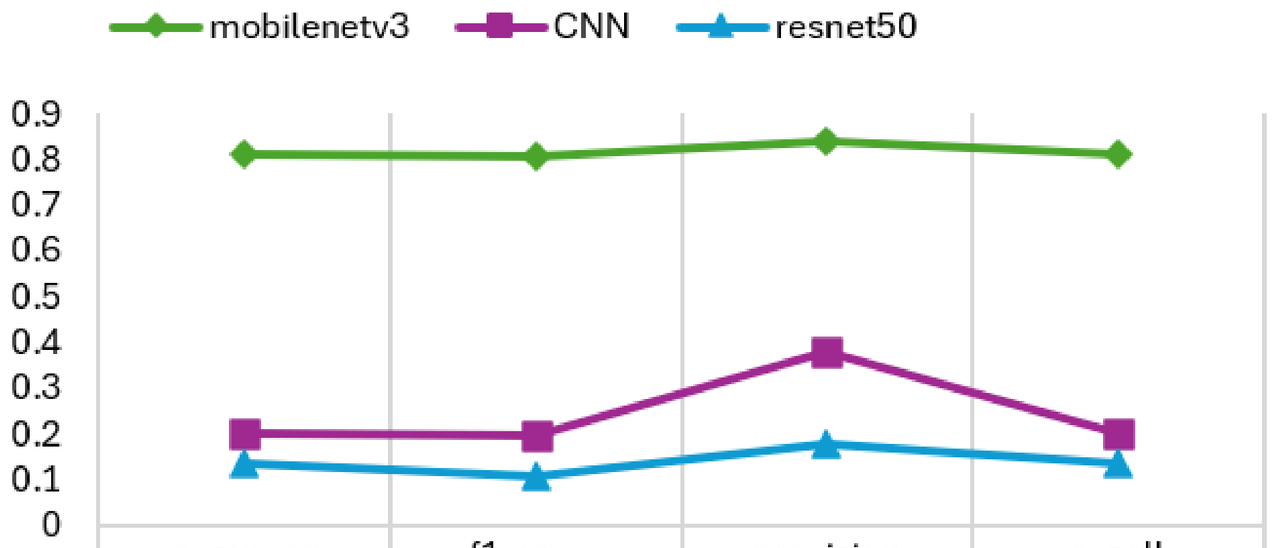
ERROR MATRIX — MOBILENETV3-LARGE



true_class	predicted_class	count
(า)210	(า)229	400
(ไ)228	(โ)226	252
(ฏ)174	(ฏ)175	244
(ฌ)172	(ณ)179	180
(ฆ์)171	(ฆ)163	140
(ั้)209	(ี)213	140
(ฆ)166	(ฆ)163	112
(ั้)233	(ี)213	100
(ใ)227	(โ)226	88
(ั)236	(ี)213	80
(ฆ์)171	(ฆ์)170	80
(ุ)216	(ู)217	76
(ี)214	(ี)213	64
(ฯ)207	(ๆ)230	64
(ั)236	(ิ)212	60
(่)232	(ี)213	60
(ิ)212	(ี)213	56
(ั้)233	(๔)245	52
(ี)215	(็)231	48
(ใ)227	(ร)195	48

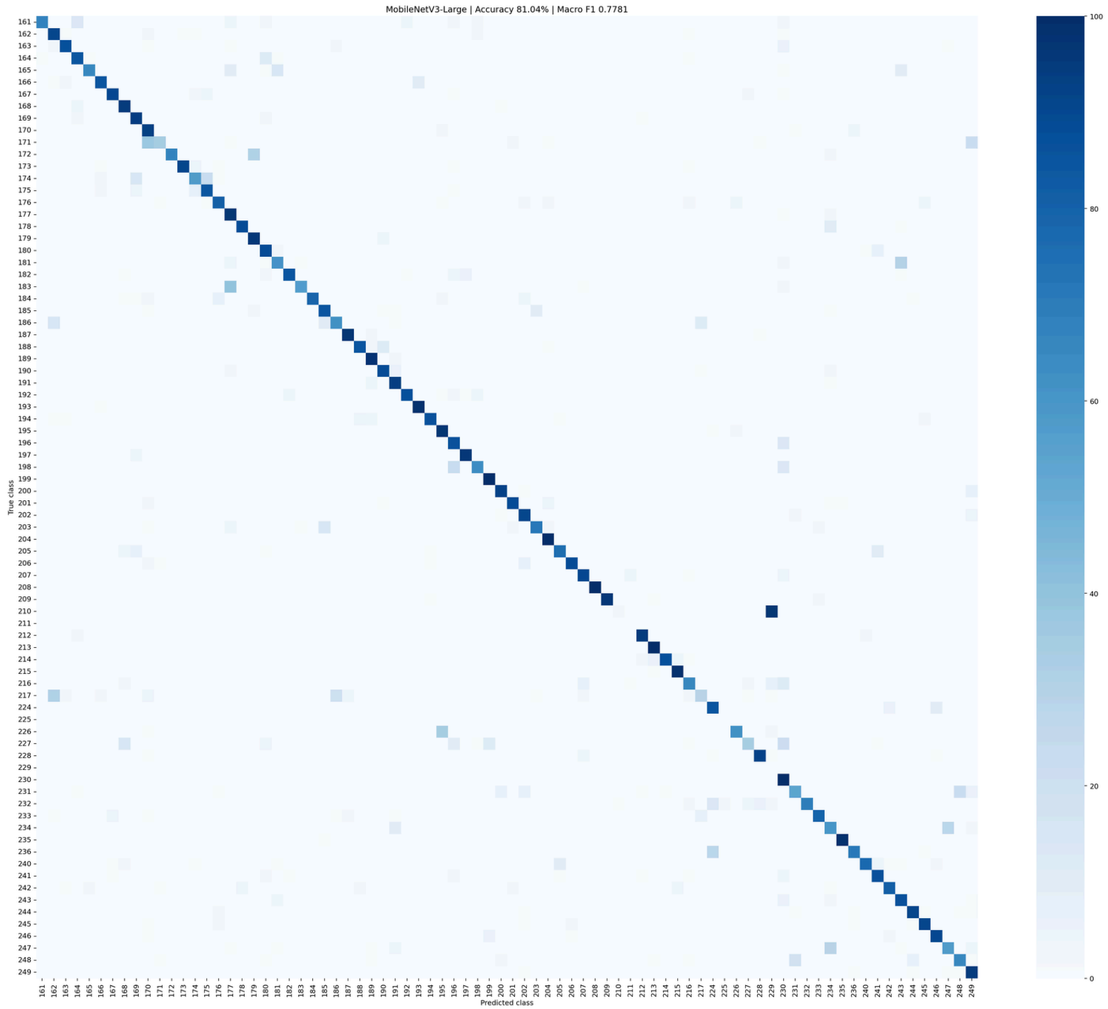
EVALUATE

ALICE EVALUATION RESULTS



	accuracy	f1-score	precision	recall
mobilenetv3	0.8104	0.8084	0.8397	0.8104
CNN	0.2003	0.1943	0.3801	0.2003
resnet50	0.1349	0.1048	0.1785	0.1349

ERROR MATRIX — MOBILENETV3-LARGE



true_class	predicted_class	count
(า) 210	(า) 229	97
(ท) 183	(ฑ) 177	40
(ช) 171	(ช) 170	38
(โ) 226	(ร) 195	34
(ฌ) 172	(ณ) 179	31
(ู) 217	(ข) 162	31
(ต) 181	(๓) 243	30
(๗) 247	(ี) 234	29
(ิ) 236	(เ) 224	28
(ี) 234	(๗) 247	28
(ภ) 198	(ฤ) 196	23
(ฎ) 174	(ฏ) 175	23
(็) 231	(๘) 248	22
(ช) 171	(๙) 249	22
(ใ) 227	(ๆ) 230	21
(ู) 217	(บ) 186	20
(๘) 248	(็) 231	19
(บ) 186	(ข) 162	16
(ค) 165	(ต) 181	15
(ฎ) 174	(ฉ) 169	15

Q&A